MALAFORGE · LIBRARY DESIGN

Sequencing library construct & barcoding

The full molecule produced by the two-step tailed-amplicon workflow, showing exactly which regions are **native to this panel's design** and which are **standard external sequences** (universal linkers & platform adapters) — and the confirmation that the sample barcodes do not conflict with any primer, tag, or the target genomes.

THE CONSTRUCT (ONE SEQUENCED MOLECULE, TOP STRAND 5′→3′)

platform adapter — external (kit) · sample barcode — native (this panel)

CS1 / CS2 linker — external (Fluidigm) · gene primer — native · amplicon insert — native target

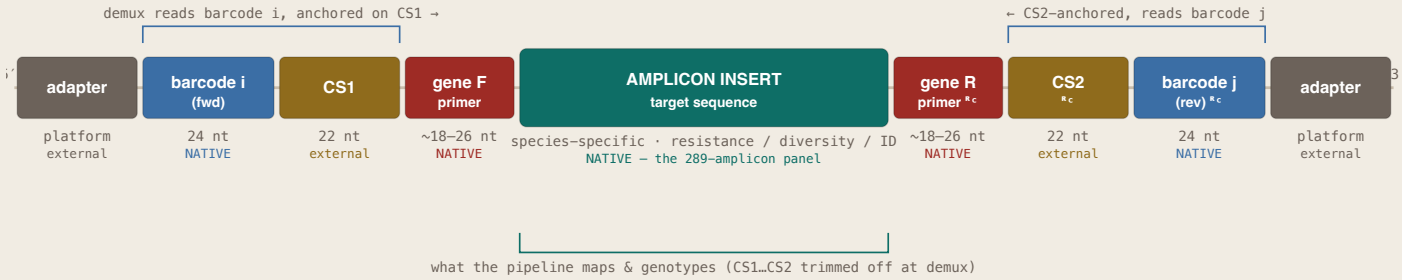


Figure. One sequenced molecule. Round-1 PCR builds **CS1**–[gene primers + insert]–**CS2**; round-2 barcoding PCR anneals to the CS1/CS2 linkers and appends the **sample barcodes** + platform adapters. Reverse-side elements are shown reverse-complemented (Rc) as they appear on the top strand.

REGION-BY-REGION — LENGTH, ORIGIN, ROLE

REGION	~LENGTH	ORIGIN	ROLE / NOTES
Platform adapter (5′ & 3′)	kit-defined	**External**	ONT / Illumina / MGI sequencing adapter, added in round-2 (or ligation). Not part of the read insert.
Sample barcode i / j	24 nt each	**Native (this panel)**	Combinatorial dual index (96 × 96 = 9,216). Set: `refs/barcodes/barcodes_96.tsv`. Read at demux.
CS1 / CS2 universal linker	22 nt each	**External (Fluidigm)**	`CS1 ACACTGACGACATGGTTCTACA` · `CS2 TACGGTAGCAGAGACTTGGTCT`
Gene-specific primers F / R	~18–26 nt	**Native**	The 289-amplicon panel; 3′ ends drive specificity (in-silico validated). CS1/CS2 are 5′ tails.
Amplicon insert	≤400 bp core	**Native target**	Species ID, resistance, diagnostic/CNV, diversity. This is what the pipeline maps & genotypes.

TWO-STEP ASSEMBLY

Round 1 — target PCR. Gene primers carry universal tails: `F = CS1 + geneF`, `R = CS2 + geneR`. Product = **CS1**–**amplicon**–**CS2**. No barcodes or adapters yet; every amplicon now shares the same CS1/CS2 handles.

Round 2 — barcoding PCR. Primers anneal to CS1/CS2 and append the sample barcode + platform adapter: `P_fwd = adapter + barcode_i + CS1`, `P_rev = adapter + barcode_j + CS2`. Because these primers prime on the constant CS1/CS2 (not on the genome), they are panel-agnostic and add the same barcodes to every amplicon in the pool.

DESIGN CONSISTENCY — NO CONFLICT (VERIFIED)

The barcode set was checked against every other sequence in the construct and the target genomes:

- ✓ barcode set orthogonality — **min pairwise Hamming = 10** (safe for 2-mismatch demux)
- ✓ barcodes vs CS1 / CS2 — **≥ 8 nt different** (worst case) → CS-anchored demux is unambiguous
- ✓ barcodes vs all **575 panel primers** — **0** within 3 mismatches → no mispriming risk
- ✓ CS1 / CS2 in the pan-reference (6 nuclear + 6 mito + human) — **0 occurrences** (all 4 orientations) → CS tails never prime on target DNA

Re-run the check any time barcodes change: `analysis/bin/check_barcode_conflicts.py`.

What was designed where. The panel-design work fixed the **scheme** (CS1/CS2 tags, a combinatorial dual 24-nt barcode plan) and the **gene primers + targets** — those are native. The actual **barcode sequences** were finalised here as an orthogonal, conflict-checked set (`generate_barcodes.py` → `barcodes_96.tsv`); swap in your reagent barcodes for a real run and re-run the check. CS1/CS2 and the sequencing adapters are standard external sequences, deliberately chosen so the barcoding chemistry is off-the-shelf.